

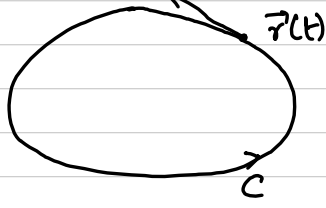
Directional derivatives, gradients & level sets

(Ref: Calculus Vol II - Tom Apostol Sec 8.15-8.16)

Defn :-

Directional derivative of $f: \mathbb{R}^n \rightarrow \mathbb{R}$ along a curve C at an arbitrary point $\vec{r}(t)$ on the curve is the directional derivative along its tangent $\vec{r}'(t)$

$$D_{\vec{b}}(f)(\vec{r}(\vec{t})) \text{ where } \vec{b} = \frac{d\vec{r}(t)}{dt}$$



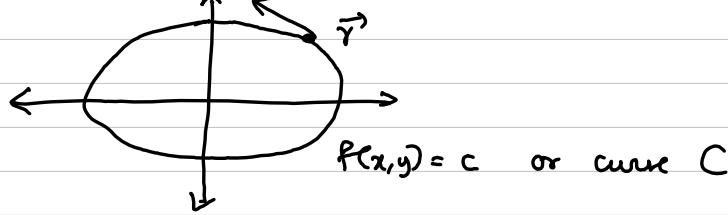
Problem 1)

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a non-constant scalar field having continuous first order partial derivatives everywhere in the plane. Let c be a constant. Then, the equation $f(x,y) = c$ describes a curve C in the plane. Assume that C has a tangent at each of its points. Prove that f has the following properties at each point of C :-

- ∇f is normal to C .
- The directional derivative of f is zero along C .
- The directional derivative of f has its largest value in a direction normal to C .

Exercises b & c

Soln :- Consider the curve $f(x,y) = c$



Let $\vec{r}(t)$ be an arbitrary point on C
 say, $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$

(i) $f(x(t), y(t)) = c \quad \forall t$

Taking derivative w.r.t t (Chain rule)

$$\begin{aligned} \frac{df}{dt} &= \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} \\ &= \nabla f(\vec{r}) \cdot \vec{r}'(t) = 0 \\ \Rightarrow (D_{\vec{r}'(t)} f)(\vec{r}(t)) &= 0 \quad \text{--- (1)} \end{aligned}$$

$\therefore$ Gradient of f at any point $\vec{r}$ in the curve has to be perpendicular to the tangent vector of C at $\vec{r}(t)$.

Thus, (a) is proved.

(ii) Recall $D_{\vec{b}}(f)(\vec{r}) = \nabla f(\vec{r}) \cdot \vec{b}$

$$\Rightarrow (D_{\vec{r}'(t)} f)(\vec{r}) = \nabla f(\vec{r}) \cdot \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|} = 0$$

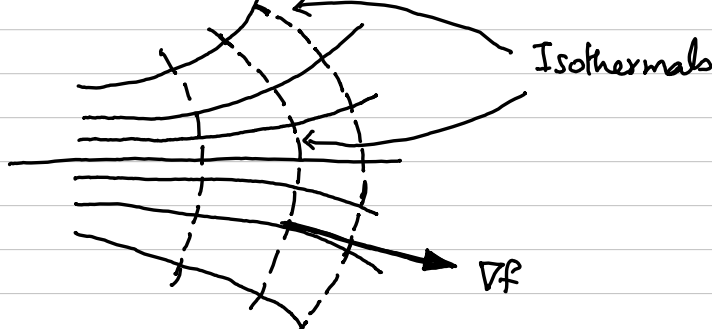
The directional derivative of f along C at $\vec{r}(t)$ is $D_{\vec{r}'(t)} f(\vec{r})$. Hence, (b) is proved.

(iii) For which $\vec{b}$ is $(D_{\vec{b}} f)(\vec{r}(t))$ the maximum?

$(D_{\vec{b}} f)(\vec{r}) = \|\nabla f(\vec{r})\|$ if $\vec{b}$ & $\nabla f(\vec{r})$ are parallel, i.e. $\vec{b}$ is along the gradient of f at $\vec{r}$.
 Thus, the directional derivative of f has its largest value in a direction normal to C .

Example

If $f(x,y)$ represents temperature at (x,y) , the curves of constant temperature called isothermals are the dotted curves below.



The flow of heat takes place in the direction of most rapid change in temperature. This direction is normal to the isothermals (level curves of $f(x,y)$, temperature function).

level sets & tangent planes

Let f be a scalar field defined on a set S in $\mathbb{R}^n$ & let c be a constant. Define

$$L(c) = \{ \vec{x} \in S \mid f(\vec{x}) = c \}$$

This set $L(c)$ is called a level set of f .

$n=2$: $L(c)$ is called a level curve

$n=3$: level surface.

Now, let $f: S \rightarrow \mathbb{R}$, where S , open subset of $\mathbb{R}^3$ & let f have continuous first order partial derivatives in S . Consider its level surface

$$L(c) : f(x,y,z) = c$$

Let $\vec{a} \in L(c)$ & consider a curve C which lies on $L(c)$ passing through $\vec{a}$. (Assume C can be parametrised by $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ which has tangents at each of its points. This is possible since we assume f has continuous first order partial derivatives.)

We shall prove that $\nabla f(\vec{a})$ is normal to C at $\vec{a}$

$$\text{Let } \vec{r}(t) \in C \Rightarrow f(\vec{r}(t)) = f(x(t), y(t), z(t)) = c$$

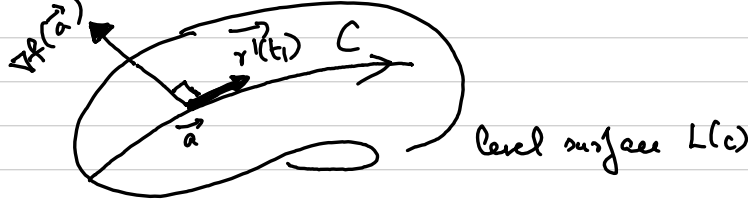
$$\Rightarrow \frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} = 0$$

$$\Rightarrow \frac{df}{dt}(\vec{r}(t)) = \nabla f(\vec{r}(t)) \cdot \vec{r}'(t) = 0$$

In particular, if $\vec{a} = \vec{r}(t_1)$, then

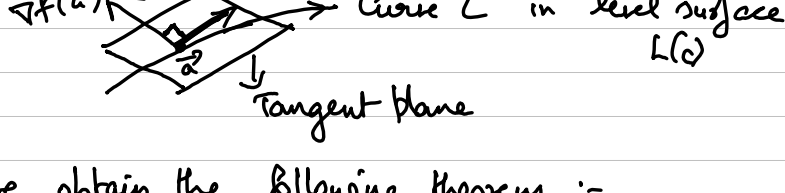
$$\nabla f(\vec{a}) \cdot \vec{r}'(t_1) = 0$$

In other words, $\text{grad}(f)$ at $\vec{a}$ is $\perp$ to the tangent vector $\vec{r}'(t_1)$ of C at $\vec{a}$.



The choice of C passing through $\vec{a}$ on the surface $L(c)$ was arbitrary. So, if we take the family of curves on $L(c)$ that pass through $\vec{a}$, then, the tangent vectors (at $\vec{a}$) $\vec{r}'(t_1)$ of all these curves are perpendicular to the gradient vector $\nabla f(\vec{a})$.

If $\nabla f(\vec{a})$ is not the zero vector, then these tangent vectors form a plane known as tangent plane to the surface $L(c)$ at $\vec{a}$.



Thus, we obtain the following theorem :-

Theorem

Let f be a scalar field $f: S \rightarrow \mathbb{R}$ where S is an open subset of $\mathbb{R}^3$ such that f has continuous first order partial derivatives on S . If the gradient of f at a point P lying on $L(c)$ ($\vec{OP} = \vec{a}$) is not the zero vector, then it is a normal vector of $L(c)$ at P .

Equation of the tangent plane at $\vec{a}$

Any point (X,Y,Z) on the tangent plane at $\vec{a}$ satisfies

$$\nabla f(\vec{a}) \cdot (X-a_1, Y-a_2, Z-a_3) = 0$$

$$\text{or } \boxed{\frac{\partial f}{\partial x}(\vec{a})(X-a_1) + \frac{\partial f}{\partial y}(\vec{a})(Y-a_2) + \frac{\partial f}{\partial z}(\vec{a})(Z-a_3) = 0}$$